Task 2 Cybersecurity and Ethical Hacking Internship

NETWORK SECURITY & SCANNING REPORT

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Course: Task-2 Network Security & Scanning
Duration: Days 13–24

PART 1 – RECONNAISSANCE

Objective

To gather information about the target system (Metasploitable) using passive and active reconnaissance techniques.

1.1 Passive Reconnaissance

Passive reconnaissance means collecting information without directly interacting aggressively with the target.

In this phase, passive information gathering techniques were used to identify network details of the target machine. The IP address of the Metasploitable machine was obtained using the ifconfig command. DNS and Whois queries were also performed (where applicable) to gather additional information.

Step 1: Find Target IP Address

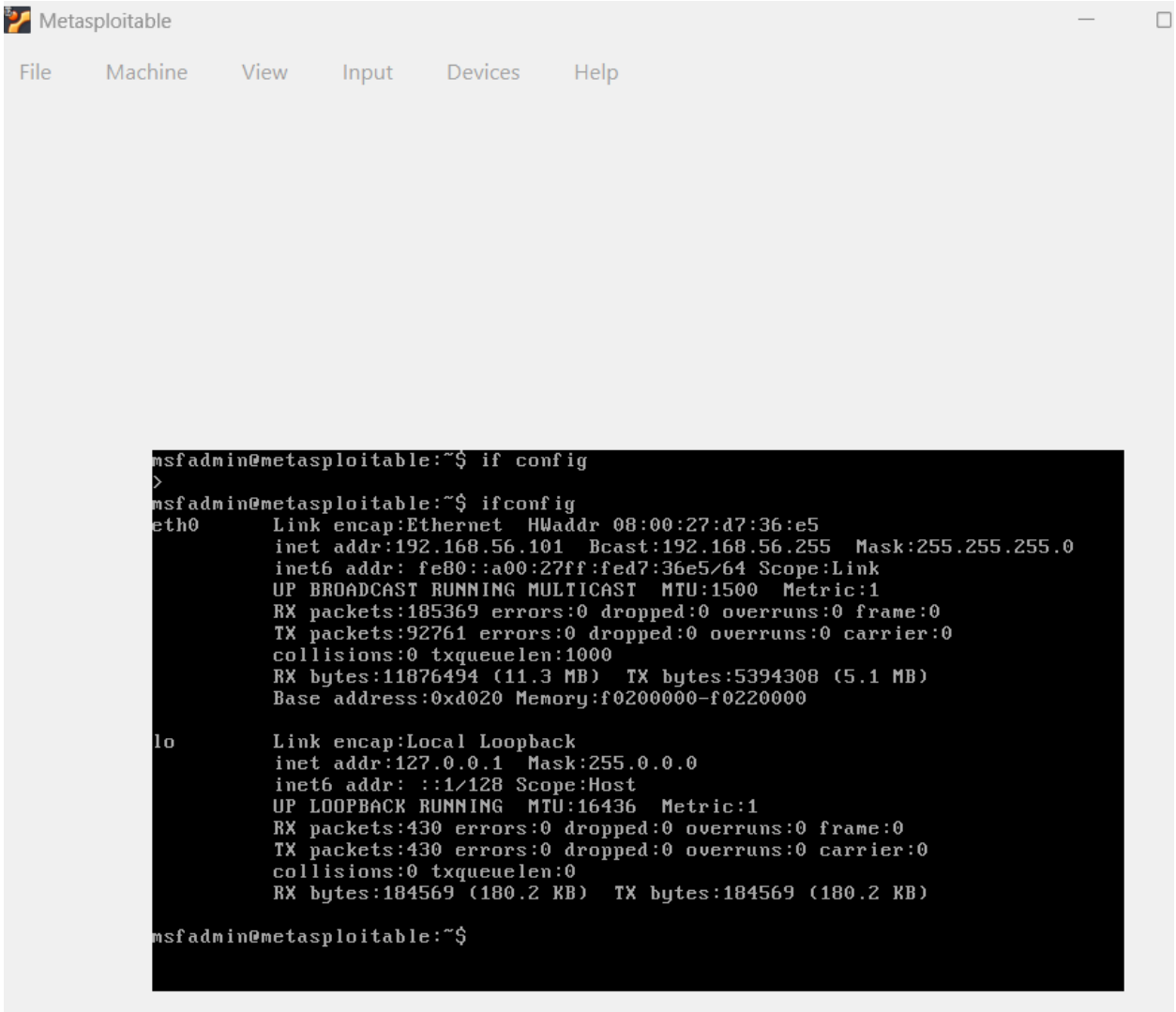


Figure 1: IP address of Metasploitable machine identified using ifconfig.

STEP 2: WHOIS Lookup

```
┌──(kali㉿kali)-[~]
└─$ whois example.com
   Domain Name: EXAMPLE.COM
   Registry Domain ID: 2336799_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.iana.org
   Registrar URL: http://res-dom.iana.org
   Updated Date: 2026-01-16T18:26:50Z
   Creation Date: 1995-08-14T04:00:00Z
   Registry Expiry Date: 2026-08-13T04:00:00Z
   Registrar: RESERVED-Internet Assigned Numbers Authority
   Registrar IANA ID: 376
   Registrar Abuse Contact Email:
   Registrar Abuse Contact Phone:
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: ELLIOTT.NS.CLOUDFLARE.COM
   Name Server: HERA.NS.CLOUDFLARE.COM
   DNSSEC: signedDelegation
   DNSSEC DS Data: 2371 13 2 C988EC423E3880EB8DD8A46FE06CA230EE23F35B578D64E78B29C3E1C83D245A
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-02-24T03:00:30Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
```

Figure 2: Whois information showing domain registration details.

Step 3: DNS Lookup

```
┌──(kali㉿kali)-[~]
└─$ nslookup example.com
Server:         172.18.222.235
Address:        172.18.222.235#53

Non-authoritative answer:
Name:   example.com
Address: 104.18.26.120
Name:   example.com
Address: 104.18.27.120
Name:   example.com
Address: 2606:4700::6812:1a78
Name:   example.com
Address: 2606:4700::6812:1b78
```

Figure 3: DNS lookup results for *example.com* using the `nslookup` command, showing the resolved IPv4 and IPv6 addresses of the domain.

1.2 Active Reconnaissance

Now we directly interact with the target.

Active reconnaissance was performed using ping and Nmap scanning techniques. The ping command confirmed that the target host was alive and reachable. A basic Nmap scan identified multiple open ports on the system, indicating exposed services running on the target machine.

Step 4: Ping Sweep

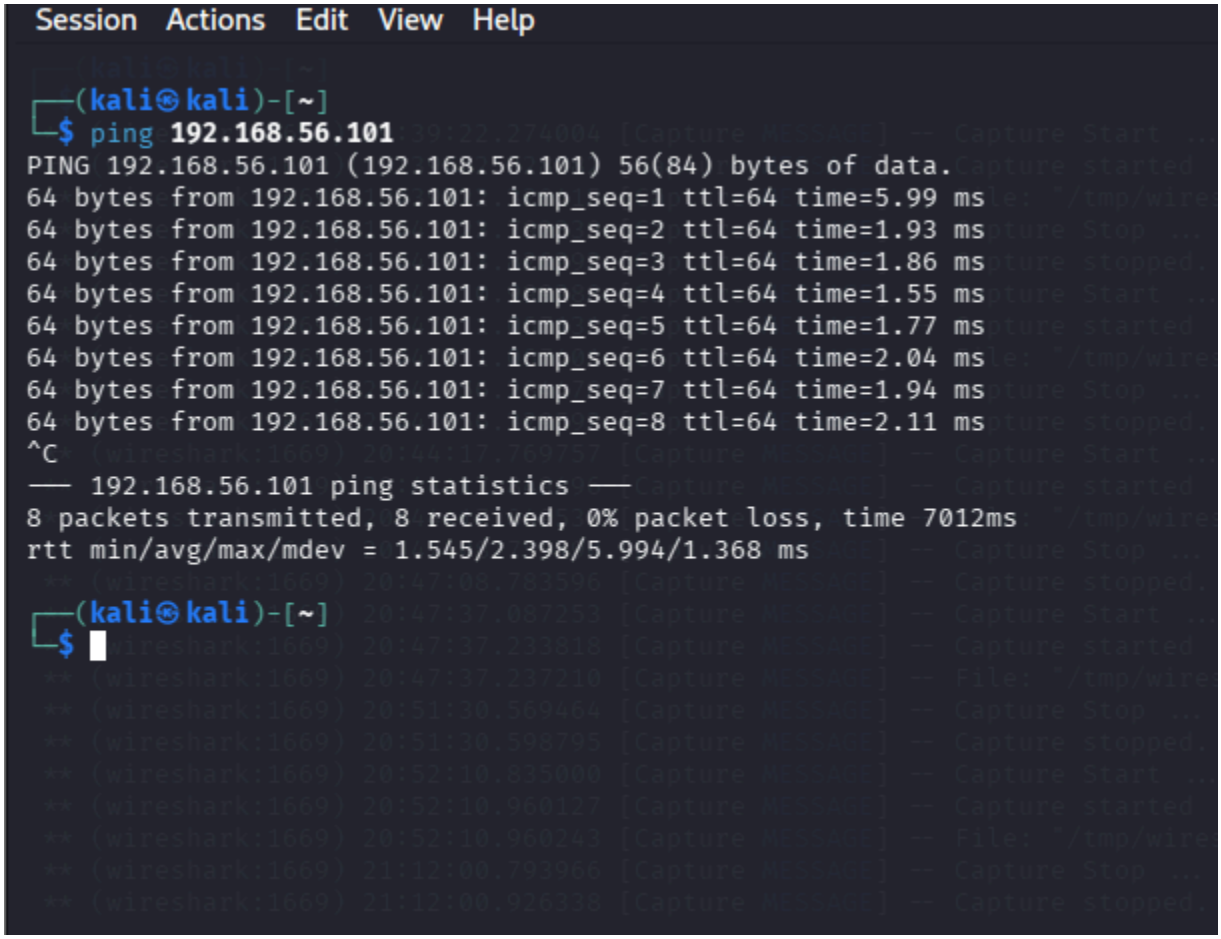


Figure 4: Ping results confirming that the target host is live.

Step 5: Basic Nmap Scan

```
└─$ nmap 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 21:51 EST
Nmap scan report for 192.168.56.101
Host is up (0.00045s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE    SERVICE
21/tcp   open     ftp
22/tcp   open     ssh
23/tcp   open     telnet
25/tcp   open     smtp
53/tcp   open     domain
80/tcp   filtered http
111/tcp  open     rpcbind
139/tcp  open     netbios-ssn
445/tcp  open     microsoft-ds
512/tcp  open     exec
513/tcp  open     login
514/tcp  open     shell
1099/tcp open     rmiregistry
1524/tcp open     ingreslock
2049/tcp open     nfs
2121/tcp open     ccproxy-ftp
3306/tcp open     mysql
5432/tcp open     postgresql
5900/tcp open     vnc
6000/tcp open     X11
6667/tcp open     irc
8009/tcp open     ajp13
8180/tcp open     unknown
MAC Address: 08:00:27:D7:36:E5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 7.97 seconds
```

Figure 5: Basic Nmap scan identifying open ports on the target system.

PART 2 – PORT & SERVICE SCANNING

Objective

To identify open ports, running services, service versions, and operating system details of the target machine using Nmap.

2.1 TCP SYN Scan

A TCP SYN scan (-sS) was performed to identify open ports on the target machine. This stealth scanning technique sends SYN packets and analyzes responses without completing the TCP handshake. Multiple ports were found open, indicating active services running on the system.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 22:12 EST
Nmap scan report for 192.168.56.101
Host is up (0.00043s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE    SERVICE
21/tcp   open     ftp
22/tcp   open     ssh
23/tcp   open     telnet
25/tcp   open     smtp
53/tcp   open     domain
80/tcp   filtered http
111/tcp  open     rpcbind
139/tcp  open     netbios-ssn
445/tcp  open     microsoft-ds
512/tcp  open     exec
513/tcp  open     login
514/tcp  open     shell
1099/tcp open     rmiregistry
1524/tcp open     ingreslock
2049/tcp open     nfs
2121/tcp open     ccproxy-ftp
3306/tcp open     mysql
5432/tcp open     postgresql
5900/tcp open     vnc
6000/tcp open     X11
6667/tcp open     irc
8009/tcp open     ajp13
8180/tcp open     unknown
MAC Address: 08:00:27:D7:36:E5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 7.98 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 6: TCP SYN scan results identifying open ports on the target system.

2.2 UDP Scan

A UDP scan (-sU) was conducted to detect open UDP services. UDP scanning is slower than TCP scanning due to lack of handshake response mechanisms.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sU --top-ports 100 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 22:31 EST
Stats: 0:00:19 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 22.22% done; ETC: 22:32 (0:00:46 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0024s latency).
Not shown: 93 closed udp ports (port-unreach)
PORT     STATE         SERVICE
53/udp   open          domain
68/udp   open|filtered dhcpc
69/udp   open|filtered tftp
111/udp  open          rpcbind
137/udp  open          netbios-ns
138/udp  open|filtered netbios-dgm
2049/udp open          nfs
MAC Address: 08:00:27:D7:36:E5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 121.16 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 7: UDP scan results (`-sU --top-ports 100`) identifying open and open|filtered UDP ports on the target system.

2.3 Service Version Detection

The -sV flag was used to identify versions of services running on open ports. This helps in determining potential vulnerabilities based on outdated software versions.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-20 22:44 EST
Nmap scan report for 192.168.56.101
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:D7:36:E5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux
_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.07 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 8: Service version detection results showing specific software versions running on open ports.

2.4 OS Detection

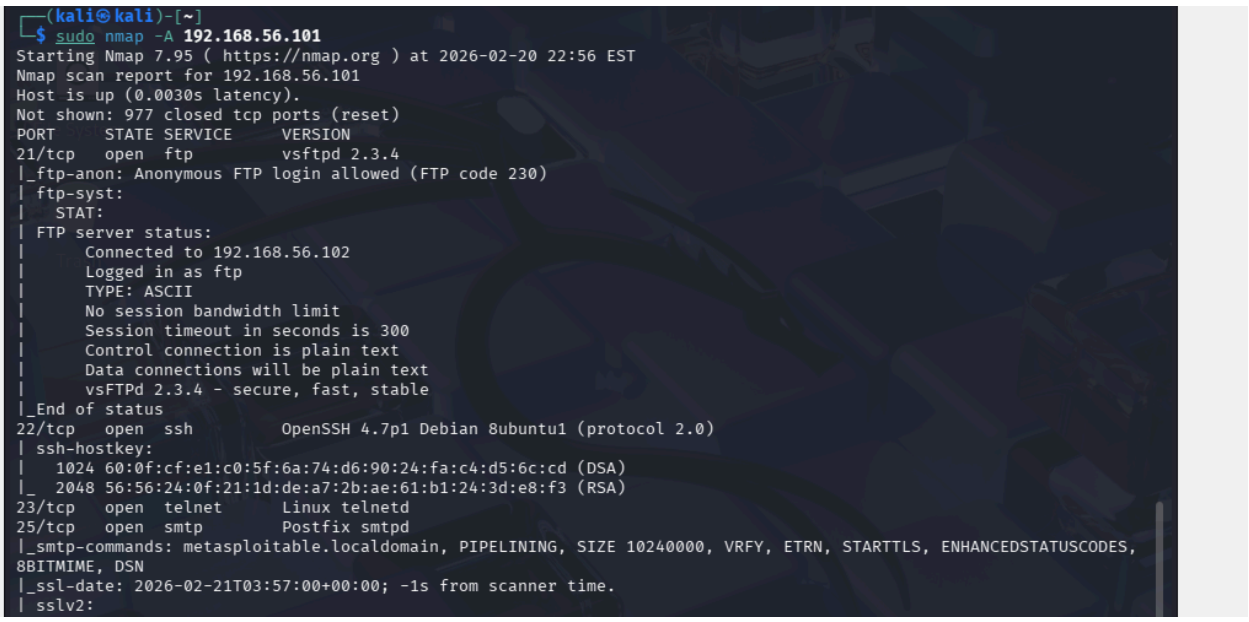
Nmap OS detection (-O) was used to identify the operating system of the target. The scan indicated that the target is running a Linux-based system, which aligns with the Metasploitable environment.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-20 22:49 EST
Nmap scan report for 192.168.56.101
Host is up (0.0028s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:D7:36:E5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.26 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 9: Operating system detection results using Nmap OS fingerprinting.

2.5 Aggressive Scan

An aggressive scan (-A) was executed to perform comprehensive enumeration of the target system. It revealed detailed service information, possible vulnerabilities, and system fingerprinting results.



```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -A 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-20 22:56 EST
Nmap scan report for 192.168.56.101
Host is up (0.0030s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.56.102
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES,
8BITMIME, DSN
|_ssl-date: 2026-02-21T03:57:00+00:00; -1s from scanner time.
| sslv2:
```

Figure 10: Aggressive Nmap scan combining OS detection, version detection, and script scanning.

PART 3 – VULNERABILITY SCANNING (Nessus Essentials)

Objective

To identify security vulnerabilities present in the target system (Metasploitable) using Nessus vulnerability scanner.

In this phase, Nessus Essentials was used to perform a vulnerability assessment on the Metasploitable target system.
Nessus is an industry-standard vulnerability scanner developed by Tenable that identifies security weaknesses and misconfigurations in systems.
The scan was configured to analyze the target IP address 192.168.56.101 and categorize vulnerabilities based on severity levels such as Critical, High, Medium, Low, and Informational.

STEP 1 – Start Nessus Service

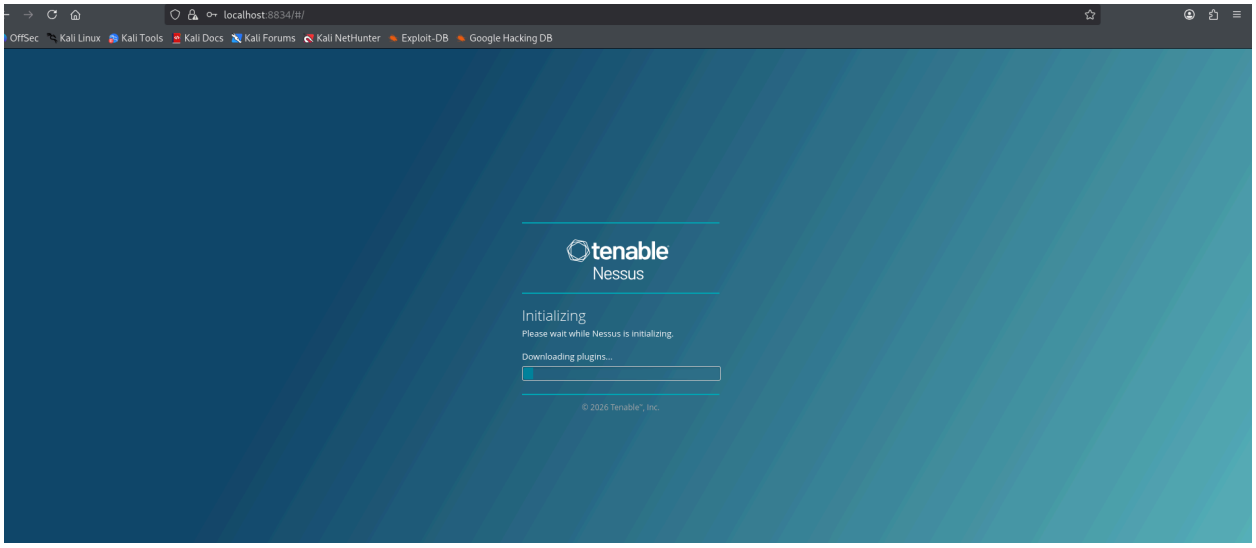


Figure 11: Nessus dashboard interface

STEP 2 – Launch Scan

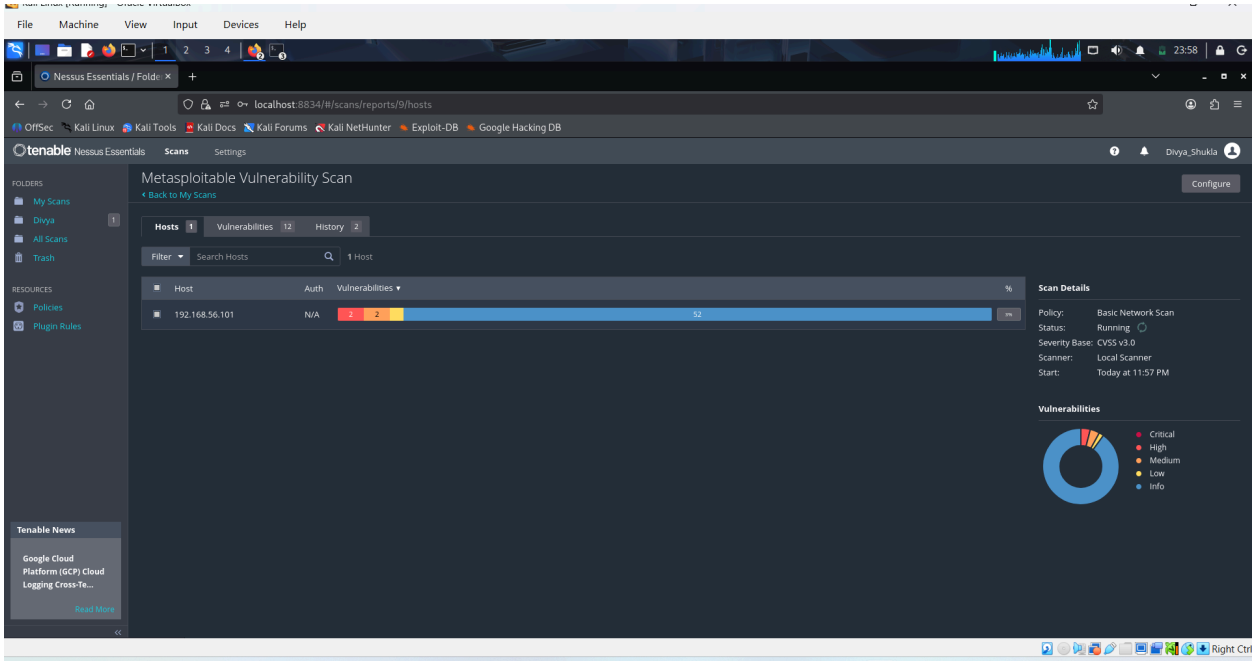


Figure 12 : Running vulnerability scan on Metasploitable using Nessus Essentials.

STEP 3 – Scan Completed

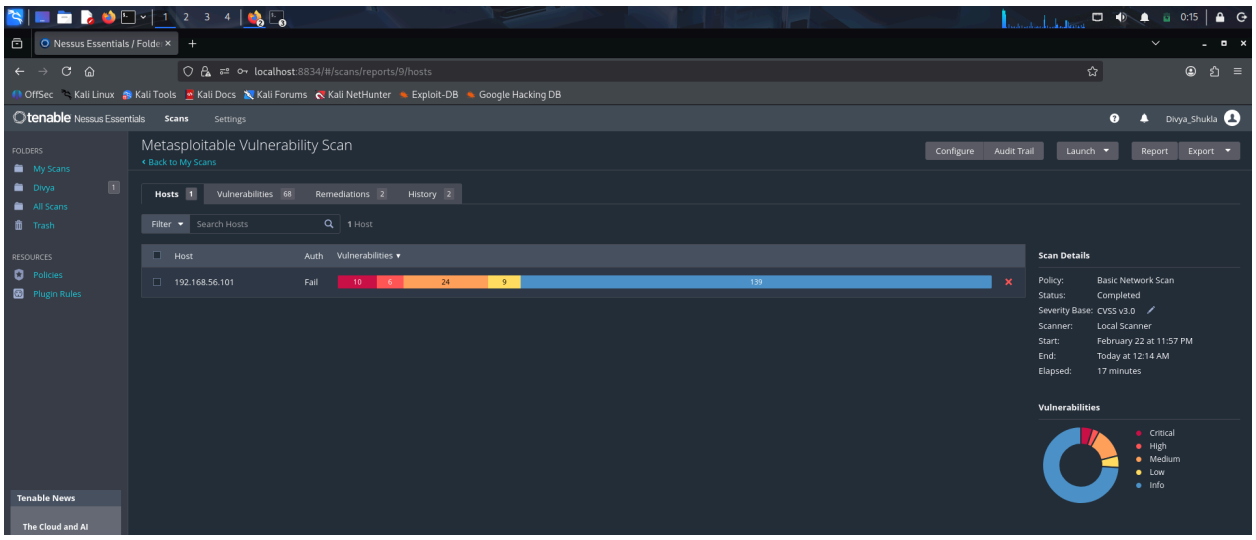


Figure 13: Nessus Basic Network Scan Results for Metasploitable Target

The screenshot shows the completed vulnerability scan performed using Nessus Essentials on the Metasploitable target (192.168.56.101). The scan identified multiple security vulnerabilities categorized by severity levels including Critical, High, Medium, Low, and Informational. The scan was completed in 17 minutes using the Basic Network Scan policy.

Detected Vulnerabilities:

- 🔴 Critical: 10
- 🟠 High: 6
- 🟡 Medium: 24
- 🟢 Low: 9
- 🔵 Informational: 139

Total vulnerabilities detected: 188

STEP 4 – View Results

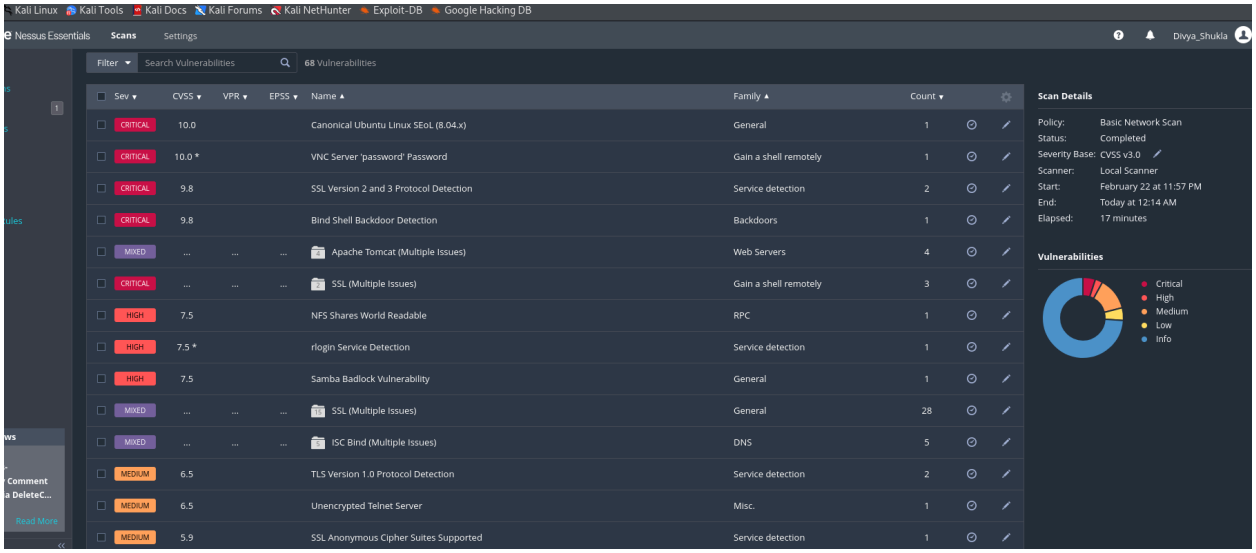


Figure 14: Vulnerability summary showing distribution of critical, high, medium, and low severity issues detected on the target system.

STEP 5 – View Detailed Vulnerability

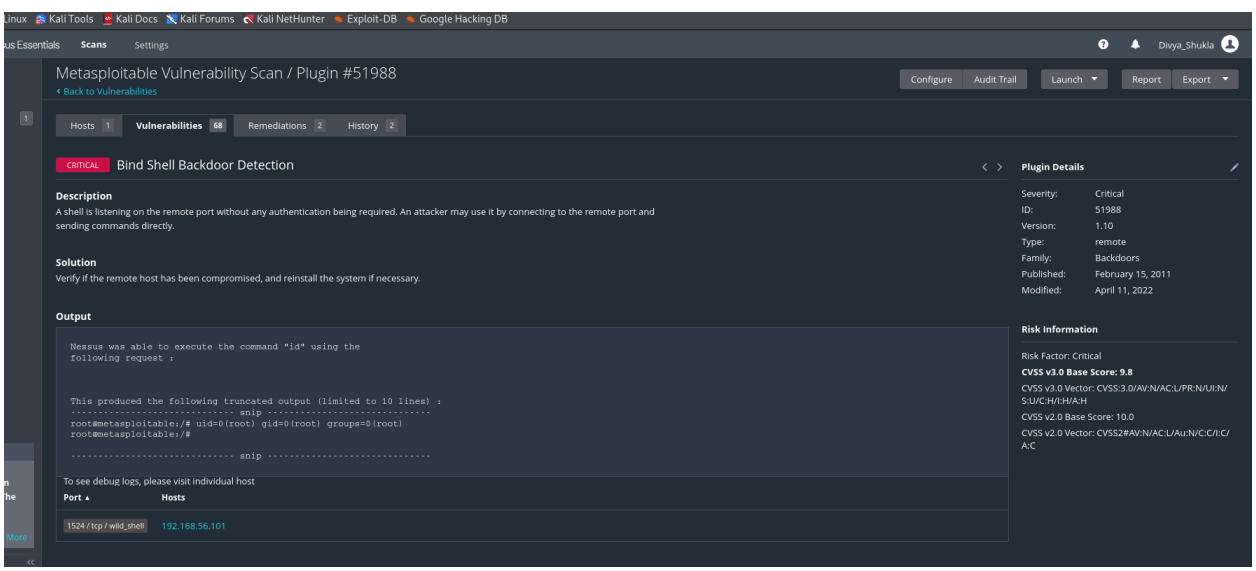


Figure 15: Critical Vulnerability – Bind Shell Backdoor Detection Identified by Nessus

PART 4 – PACKET ANALYSIS USING WIRESHARK

Objective

To capture and analyze network traffic between Kali Linux and Metasploitable using Wireshark.

STEP 1 – Start Wireshark

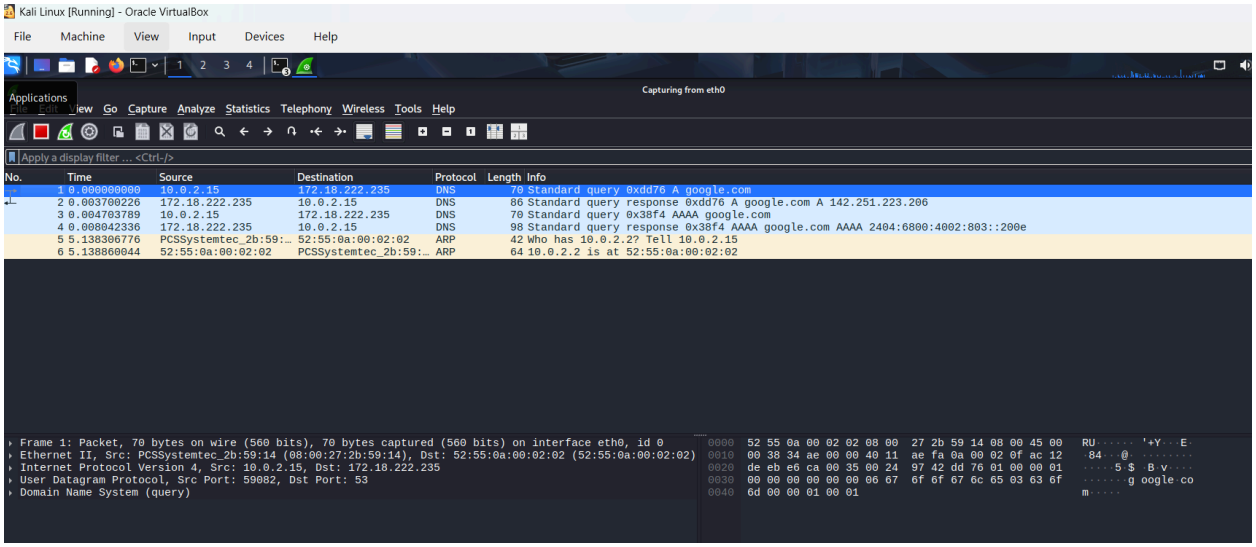


Figure 16: Wireshark capturing live network traffic on the selected network interface.

STEP 2 – Capture FTP Traffic

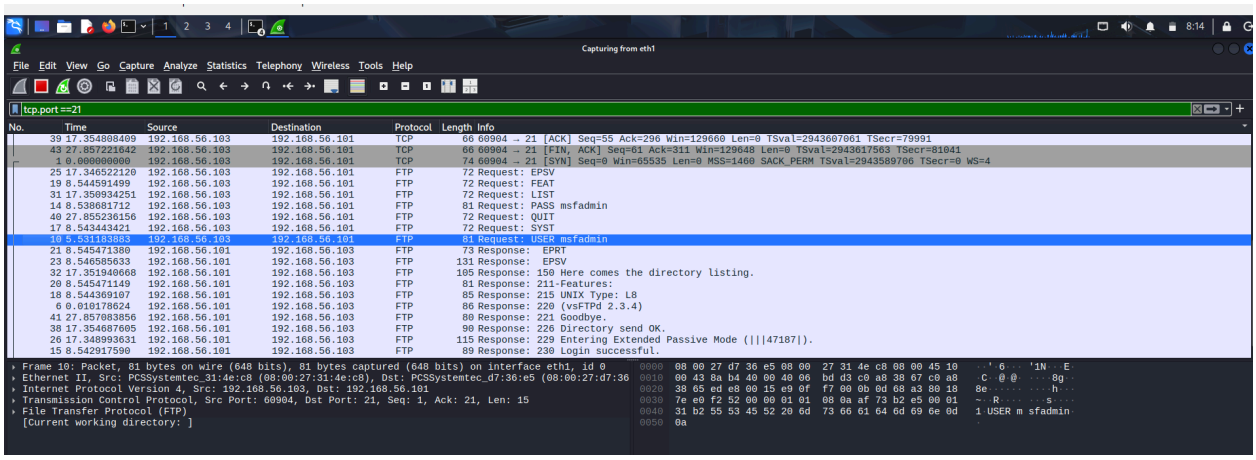


Figure 17: Wireshark capture showing unencrypted FTP login credentials (USER msfadmin / PASS msfadmin) transmitted over TCP port 21

STEP 3 – Capture HTTP Traffic

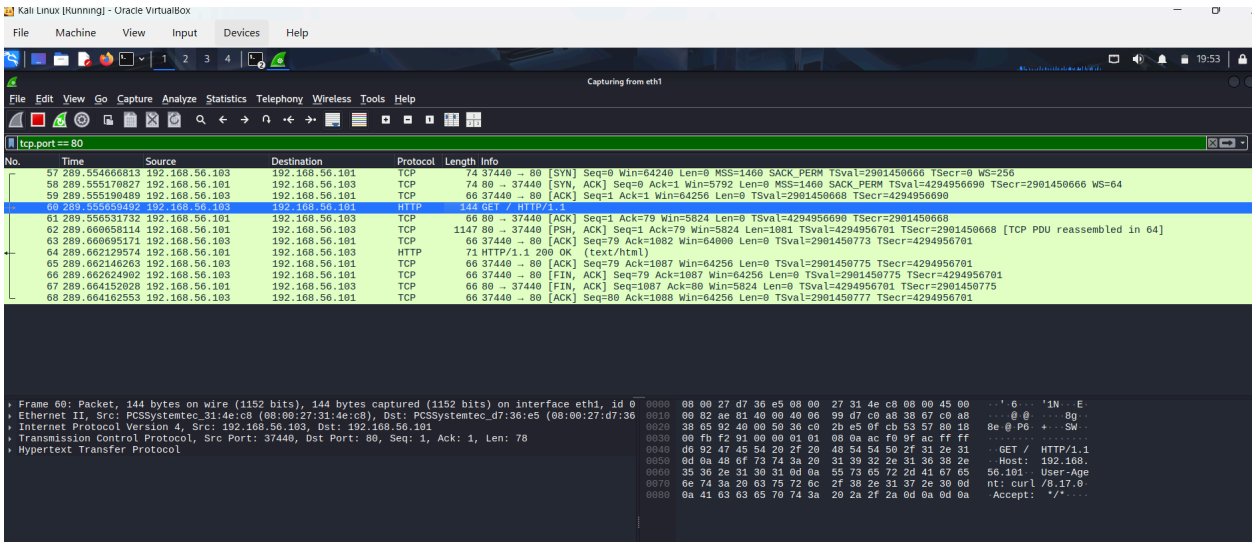


Figure 18: Wireshark capture showing HTTP GET request and server response over TCP port 80

STEP 4 – Capture DNS Traffic

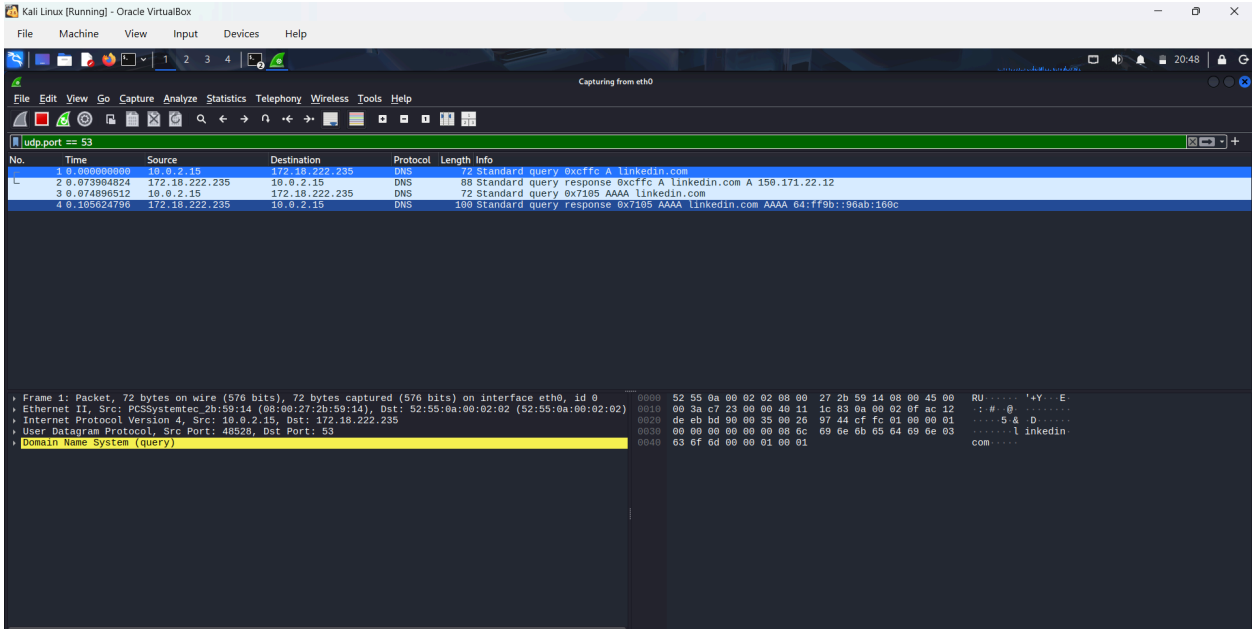


Figure 19: Wireshark capture showing DNS standard query and response for linkedin.com over UDP port 53

STEP -5 Filter credentials from unencrypted FTP traffic

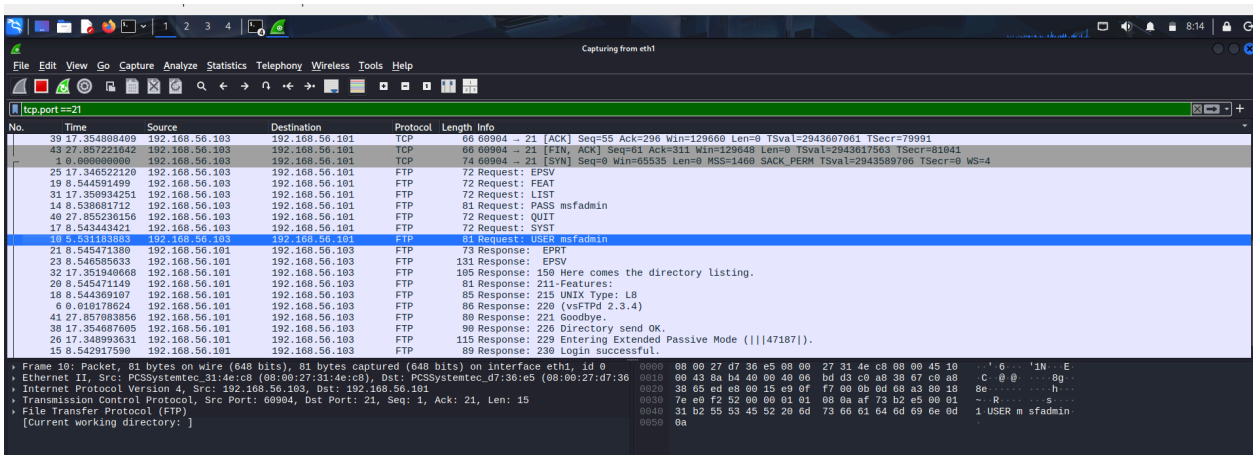


Figure 20: Wireshark capture filtered using `tcp.port == 21`, revealing FTP login credentials (`USER msfadmin` and `PASS msfadmin`) transmitted in plaintext between Kali

(192.168.56.103) and Metasploitable (192.168.56.101).

STEP 6 -TCP SYN Flood

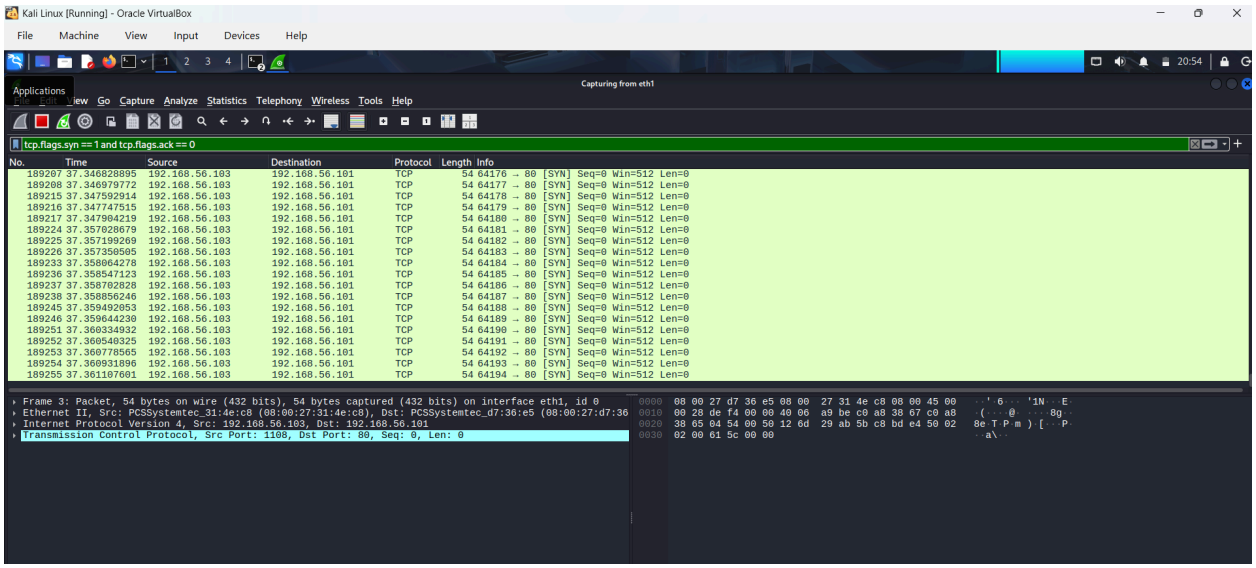


Figure 21: Wireshark capture of TCP SYN flood attack showing multiple SYN packets from Kali (192.168.56.103) to Metasploitable (192.168.56.101) on port 80

PART 5 – FIREWALL BASICS

Objective

To configure basic firewall rules using `iptables` to block specific ports and prevent unauthorized access.

STEP 1 – Check Current Firewall Rules

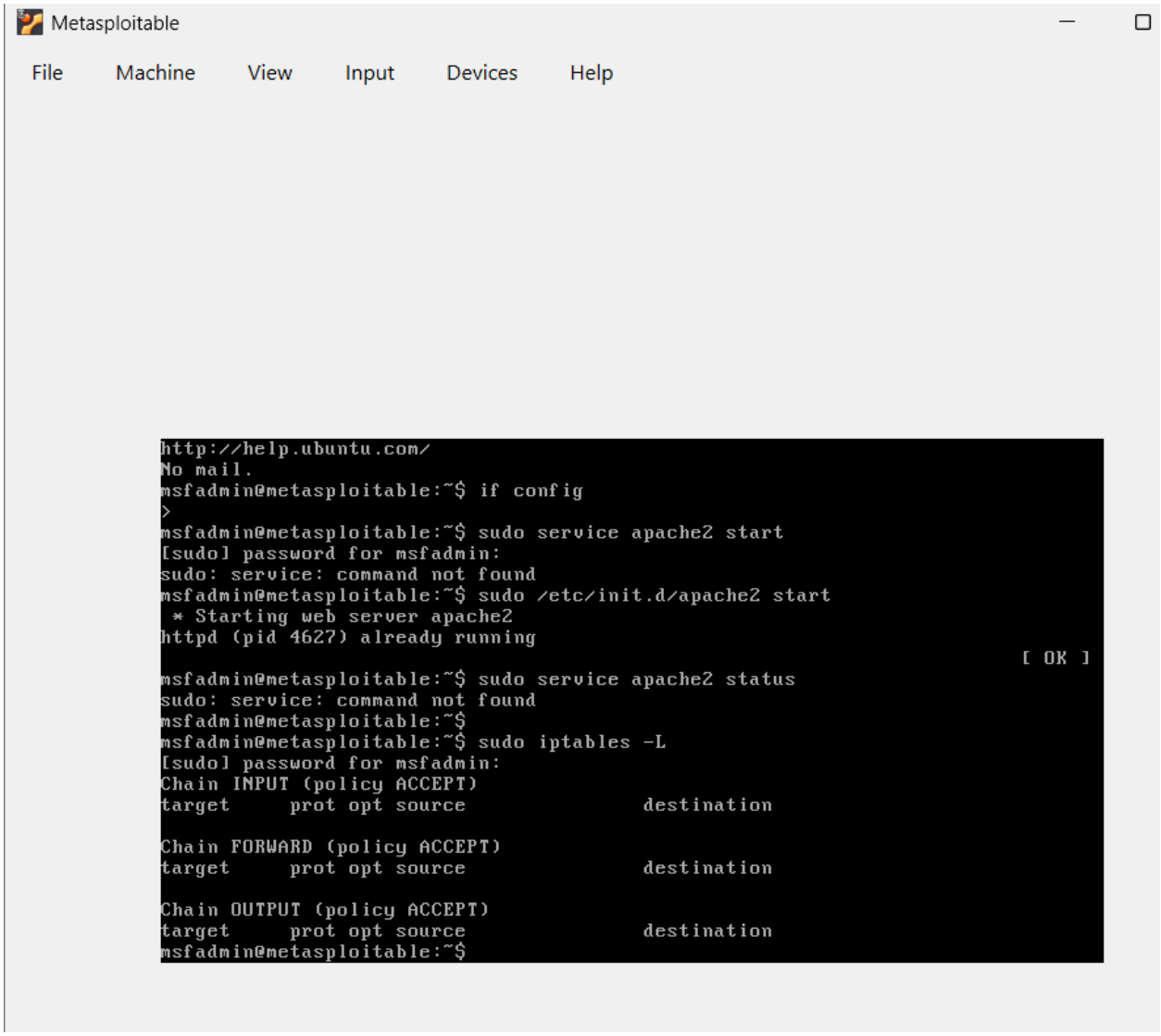


Figure 22: Initial iptables configuration on Metasploitable with default ACCEPT policies before applying firewall rules

STEP 2 – Verify Rule

```
msfadmin@metasploitable:~$ sudo service apache2 status
sudo: service: command not found
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ sudo iptables -L
[sudo] password for msfadmin:
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
msfadmin@metasploitable:~$ sudo iptables -A INPUT -p  tcp --dport 80 -j DROP
msfadmin@metasploitable:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       tcp  --  anywhere             anywhere            tcp dpt:www

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
msfadmin@metasploitable:~$ _
```

Figure 23: Updated iptables configuration showing DROP rule applied to TCP destination port 80

STEP 3 – Test Block From Kali

```
┌──(kali㉿kali)-[~]
└─$ nmap -p 80 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 21:13 EST
Nmap scan report for 192.168.56.101
Host is up (0.0026s latency).

PORT   STATE    SERVICE
80/tcp filtered http
MAC Address: 08:00:27:D7:36:E5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 6.98 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 24: Nmap scan result showing port 80 filtered after applying iptables DROP rule